Dave Morrison was visiting me at Cornell during the spring term of 1995, and that afternoon we got together to discuss Strominger's paper. Within a couple of hours we had an outline of what the "second half of the story" might look like. Drawing on some insights from the late 1980s of the mathematicians Herb Clemens of the University of Utah, Robert Friedman of Columbia University, and Miles Reid of the University of Warwick, as applied by Candelas, Green, and Tristan Hübsch, then of the University of Texas at Austin, we realized that when a three-dimensional sphere collapses, it may be possible for the Calabi-Yau space to tear and subsequently repair itself by reinflating the sphere. But there is an important surprise. Whereas the sphere that collapsed had three dimensions, the one that reinflates has only two. It's hard to picture what this looks like, but we can get an idea by focusing on a lower-dimensional analogy. Rather than the hard-to-picture case of a three-dimensional sphere collapsing and being replaced by a two-dimensional sphere, let's imagine a one-dimensional sphere collapsing and being replaced by a zero-dimensional sphere.

First of all, what are one- and zero-dimensional spheres? Well, let's reason by analogy. A two-dimensional sphere is the collection of points in three-dimensional space that are the same distance from a chosen center, as shown in Figure 13.2(a). By following the same idea, a one-dimensional sphere is the collection of points in two-dimensional space (the surface of this page, for example) that are the same distance from a chosen center. As shown in Figure 13.2(b), this is nothing but a circle. Finally, following the pattern, a zero-dimensional sphere is the collection of points in a one-dimensional space (a line) that are the same distance from a chosen center. As shown in Figure 13.2(c), this amounts to two points, with the "radius" of the zero-dimensional sphere equal to the distance each point is from their common center. And so, the lower-dimensional analogy alluded to in the preceding paragraph involves a circle (a one-dimensional sphere) pinching down, followed by space tearing, and then being replaced by a zero-dimensional sphere (two points). Figure 13.3 puts this abstract idea into practice.

We imagine beginning with the surface of a doughnut, in which a one-dimensional sphere (a circle) is embedded, as highlighted in Figure 13.3. Now, let's imagine that as time goes by, the highlighted